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Study Notes

Revision Booklet - Finance

Financial System & Financial Market

Financial System

A financial system is a network of institutions, markets, instruments, and services that facilitate the transfer of funds within an economy. It plays a crucial role in economic growth and development by efficiently allocating resources, promoting investment, and facilitating trade and commerce.

Role of Financial Regulators and key Regulatory bodies

Financial regulators play a critical role in maintaining the integrity and stability of the financial system. Their primary responsibilities include regulation, supervision, and enforcement of rules and standards to protect the interests of the public and ensure smooth functioning of the financial markets and institutions.

Regulator	Sector	Establishment	Headquarters	Functions
RBI	Banking, Money Market	April 1, 1935	Mumbai	Monetary policy, financial supervision, foreign exchange management, currency issuance
SEBI	Capital Markets	April 12, 1992	Mumbai	Regulating securities markets, protecting investors
IRDAI	Insurance	April 2000	Hyderabad	Regulating insurance and re-insurance industries
PFRDA	Pension	February 2014	New Delhi	Regulating pension funds
IFSCA	IFSCs	April 27, 2020	GIFT City, Gandhinagar	Unified regulation of financial products in IFSCs
NABARD	Agriculture and Rural Dev	July 12, 1982	Mumbai	Providing credit for agriculture and rural development
NHB	Housing Finance	July 9, 1988	New Delhi	Regulation and supervision of housing finance companies

Financial Market

Financial markets are platforms for the trading of financial claims and services. Participants include corporations, financial institutions, individuals, and governments. The markets facilitate the buying and selling of assets like foreign exchange, stocks, bonds, and derivatives.

Classification of Financial Markets

1. By Nature of Claim

Type	Description	Examples
Debt Market	Financial market where investors buy and sell debt securities, mostly in the form of bonds.	Government Securities, Corporate Debt
Equity Market	Platform where companies issue shares for sale, helping them raise capital.	Stocks/Shares

2. By Structure

Type	Description	Examples
Exchange Traded Markets	Markets where standardized contracts are traded under the supervision of a stock exchange.	National Stock Exchange (NSE), Bombay Stock Exchange (BSE), New York Stock Exchange (NYSE)
Over-the-Counter (OTC) Markets	Markets where trading occurs over a computer and telephone-linked network of dealers.	Forex Market, Debt Instruments, Private Negotiations

3. By Nature of Maturity of Claims

Type	Description	Examples
Money Market	Deals in short-term claims with a maturity period of one year or less.	Treasury Bills, Call Money Market, Commercial Bills Market
Capital Market	Deals in long-term claims with a maturity period of more than one year.	Stock Markets, Government Bond Markets, Corporate Bond Markets, Derivatives Markets

4. By Timing of Delivery

Type	Description	Examples
Spot/Cash Market	Market where the buying and selling of securities are done at the same time on a cash basis.	Equity Cash Segment, T+1 Settlement Cycle
Derivatives Market	Market for financial instruments like futures and options based on the values of their underlying assets.	Futures Contracts, Options Contracts

Comparison of Primary and Secondary Markets

Feature	Primary Market	Secondary Market
Nature	New issue market	Market for existing securities
Direct/Indirect	Direct (issuer to investor)	Indirect (investor to investor)
Financing	Provides capital to companies	No direct funding to companies
Price Determination	Through book building	Market-driven
Intermediaries	Underwriters, merchant bankers	Brokers, stock exchange

Credit Rating Agencies

Credit Rating Agency

Credit rating is a quantified independent assessment of a borrower's creditworthiness, reflecting their ability and willingness to service the loan.

Role of Credit Rating Agencies (CRAs):

- Assess and assign credit ratings to borrowers or debt instruments.
- Analyze financials and repayment capabilities.
- Help lenders price loans based on credit risk.

Major Global CRAs:

CRA	Country of Origin	Establishment Year	Rating Scale
Standard & Poor's	USA	1860	AAA, AA, A, BBB, BB
Moody's	USA	1909	Aaa, Aa, A, Baa, Ba
Fitch Ratings	USA	1914	AAA, AA, A, BBB, BB

Major Indian CRAs:

CRA	Promoter	Establishment Year	Headquarters
CRISIL Ltd.	Standard & Poor's (67.53%)	1987	Mumbai
CARE Ratings Ltd.	LIC (9.85%) and PSU Banks	April 1993	Mumbai
ICRA Limited	Moody's (51.86%)	1991	New Delhi
India Ratings and Research	Fitch Group (100%)	1995	Mumbai
Brickwork Ratings	Vivek Kulkarni (68%)	2008	Bengaluru
Acuité Ratings (formerly SMERA)	SIDBI (35.73%)	September 2005	Mumbai
Infomerics Valuation and Rating	Vipin Malik (71.07%)	2015	New Delhi

Credit Rating Scale in India:

Rating Grade	Safety Level	Long Term Rating	Short Term Rating
Investment Grade	Highest	AAA	A1(+)
High	High	AA(+/-)	A2(+)

Adequate	Moderate	BBB(+/-)	A3(+)
Speculative Grade	Moderate Risk	BB(+/-)	
High Risk	High Risk	B(+/-)	A4(+)
Very High Risk	Very High Risk	C	
Default	Default	D	D

LIBOR & SOFR Transition

LIBOR (London Interbank Offered Rate):

- Benchmark rate for short-term loans between leading banks.
- Based on five currencies (USD, EUR, GBP, JPY, CHF) and seven maturities (overnight to 12 months).
- Administered by ICE Benchmark Administration (IBA).

Transition to SOFR (Secured Overnight Financing Rate):

- SOFR is a benchmark interest rate for dollar-denominated derivatives and loans, replacing LIBOR.
- Based on transactions in the Treasury repurchase market.
- Seen as more accurate due to extensive trading in the Treasury repo market.

Financial Market: Money Market & Capital Market

Money Market

1. Overview

Feature	Description
Definition	Market for short-term funds, ranging from overnight to one year.
Role	Meets short-term liquidity requirements, provides central point for central bank's intervention.
Participants	RBI, Scheduled Commercial Banks, Cooperative Banks, Corporates, Mutual Funds, DFHI.

2. Structure of Indian Money Market

Sector	Description
Organized Sector	Includes RBI, Commercial Banks, Cooperative Banks, NBFCs.
Unorganized Sector	Includes Chit Funds, Money Lenders, Nidhis, Indigenous Bankers.

3. Participants

Participant	Description
RBI	Core participant, regulates the money market, implements monetary policy.
Scheduled Commercial Banks	Major borrowers/suppliers of short-term funds, mobilize savings, provide short-term funds to government.
Cooperative Banks	Perform similar functions as commercial banks in the money market.
Corporates	Raise short-term funds for working capital requirements (e.g., through commercial papers).
Mutual Funds	Invest funds in short-term instruments to earn returns for investors.
Discount and Finance House of India (DFHI)	Deals in short-term money market instruments, participates in inter-bank call/notice money market, rediscounts T-Bills.

4. Instruments in Money Market

Instrument	Description
Call Money Market	Borrowing or lending of unsecured funds on an overnight basis.
Notice Money Market	Borrowing or lending of unsecured funds for up to 14 days.
Term Money Market	Borrowing or lending of unsecured funds for a period beyond 14 days.
Treasury Bills	Short-term debt instruments issued by the Government of India, available in tenors of 91, 182, and 364 days.
Commercial Papers	Short-term debt instruments issued by corporates to raise funds.

5. Interest Rates and Trading Venues

Feature	Description
Interest Rates	Determined by eligible participants, average interest rate in the call money market known as Weighted Average Call Rate (WACR).
Trading Venues	Executed in Over-the-Counter markets, including NDS-CALL platform or other authorized electronic trading platforms by RBI.

6. Primary Dealers

Feature	Description
Definition	Organizations registered with RBI, licensed to buy and sell securities, act as underwriters in the primary market and market makers in the secondary market.
Introduction	Primary Dealers system introduced by RBI in 1995.

Types	Most PDs are started by scheduled commercial banks and registered as NBFCs.

Comparison of Primary and Secondary Markets:

Feature	Primary Market	Secondary Market
Nature	New issue market	Market for existing securities
Direct/Indirect	Direct (issuer to investor)	Indirect (investor to investor)
Financing	Provides capital to companies	No direct funding to companies
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Capital Market

1. Overview

Feature	Description
Definition	Market for long-term debt or equity-backed securities.
Role	Facilitates raising capital, provides investment opportunities, aids in resource allocation and economic growth.

2. Classification of Capital Markets

Classification Criteria	Type	Description	Examples
By Nature of Instrument	Debt Market	Market for trading long-term debt securities.	Corporate Bonds, Government Bonds
	Equity Market	Market for trading equity instruments (shares).	Common Stock, Preferred Shares
By Market Structure	Primary Market	Market for new securities issuance.	Initial Public Offerings (IPO), Follow-on Public Offerings (FPO)
	Secondary Market	Market for trading existing securities.	Stock Exchanges, Over-the-Counter (OTC) Markets

3. Participants

Participant	Description

Companies	Issue stocks or bonds to raise capital for expansion and growth.
Investors	Individuals or institutions that invest in stocks or bonds to earn returns.
Stock Exchanges	Organized platforms for buying and selling securities.
Regulatory Bodies	Ensure fair and transparent functioning of the capital markets.
Intermediaries	Include brokers, dealers, underwriters, and advisors who facilitate transactions in the market.

4. Instruments in Capital Markets

Instrument	Description
Stocks	Equity instruments representing ownership in a company.
Bonds	Debt instruments where the issuer owes the holders a debt and is obliged to pay interest and repay principal.
Derivatives	Financial contracts whose value is derived from underlying assets like stocks, bonds, commodities.
Mutual Funds	Investment vehicles that pool funds from investors to invest in diversified portfolios of stocks, bonds, etc.

5. Stock Exchanges in India

Exchange Name	Location	Segments Permitted
Bombay Stock Exchange (BSE)	Mumbai	Equity, Equity Derivatives, Currency Derivatives, Commodity Derivatives, Debt
National Stock Exchange (NSE)	Mumbai	Equity, Equity Derivatives, Currency Derivatives, Commodity Derivatives, Debt
Multi Commodity Exchange (MCX)	Mumbai	Commodity Derivatives
India International Exchange (INX)	GIFT City, Gujarat	Equity Derivatives, Commodity Derivatives, Currency Derivatives, Debt
NSE IFSC Ltd	GIFT-SEZ, Gujarat	Equity Derivatives, Currency Derivatives, Commodity Derivatives, Debt (Masala Bonds)

6. Types of Capital Market Instruments

Instrument	Description
Equity Shares	Represent ownership in a company, entitle shareholders to voting rights and dividends.
Preference Shares	Type of equity that pays fixed dividends and has priority over common shares in the event of liquidation.
Corporate Bonds	Long-term debt instruments issued by corporations to raise capital.

Government Bonds	Long-term debt securities issued by the government to finance public projects.
Debentures	Unsecured debt instruments backed only by the creditworthiness of the issuer.
Convertible Securities	Bonds or preferred shares that can be converted into a specified number of common shares.
Derivatives	Include futures, options, and swaps, used for hedging risk or speculative purposes.

7. Regulatory Bodies

Regulator	Sector	Functions
Securities and Exchange Board of India (SEBI)	Capital Markets	Regulates securities markets, protects investors, ensures market integrity and transparency.
Reserve Bank of India (RBI)	Banking and Financial Markets	Regulates money markets, oversees monetary policy, ensures financial stability.

Alternative Investment Funds

Forfeiting, P2P Lending, Crowdfunding, Angel Investors, and Venture Capital

Forfeiting

Feature	Description
Definition	Arrangement where a forfeiter (bank or finance company) purchases a transferable negotiable instrument (Letter of Credit/Bank Guarantee) from an exporter at a discount.
Key Features	Medium-term financing, used only in international trade, always without recourse.
Differences from Factoring	Factoring can be used for domestic and international trade, may be with or without recourse, generally used for smaller amounts.

Peer-to-Peer Lending (P2P Lending)

Feature	Description
Definition	Platform that allows direct interaction between lenders and borrowers, enabling contractual agreements based on mutually agreed terms.
Key Features	Maximum reach due to online platform, uncollateralized loans, regulated by RBI as NBFC.
RBI Regulations	Minimum capital of ₹2 crore, leverage ratio max 2x, P2Ps act as intermediaries, maintain loan contract documents, max exposure of ₹10 lakh per lender/borrower, max loan tenure 36 months.

Crowdfunding

Feature	Description
Definition	Arrangement where many investors pool money to fund a project, business venture, or social cause, typically via internet/social media.
Examples	Milaap, Wishberry.

Angel Investors

Feature	Description
Definition	Equity-based financing for startups, provided by affluent individuals or consortiums in exchange for equity stakes.
Key Features	Investment horizon of 3-8 years, high-risk investment, investors expect substantial stake.

Venture Capital Funds

Feature	Description
Definition	Equity funding for startups, backed by institutional clients, demanding decision-making power in the company.
Key Features	Investment horizon of 3-5 years, larger funds compared to angel investors, help startups with little earnings.

Leasing, Franchising, and Factoring

Leasing

Type	Description
Finance Lease	Long-term lease transferring ownership to the lessee at lease end, lessee capitalizes the asset on balance sheet.
Operating Lease	Short-term lease, ownership remains with lessor, lease payments treated as operating expenses.

Other Forms	Description
Direct Lease	Lessor buys property to lease to lessee, usually offered by financing institutions.
Sale and Leaseback	Seller of asset leases it back from purchaser, typically used for buildings or large production assets.
Leveraged Lease	Lessor finances leased asset by taking a loan, lessee pays lessor, lessor pays lender.

Franchising

Feature	Description
Definition	One party gives license to another for using brand/trademark in exchange for payment (franchise fee + sales percentage).
Advantages for Franchiser	Geographic expansion without own investment, additional income from franchise fees.
Advantages for Franchisee	Reduced risk, established brand name, ease of running enterprise with well-established techniques.

Factoring

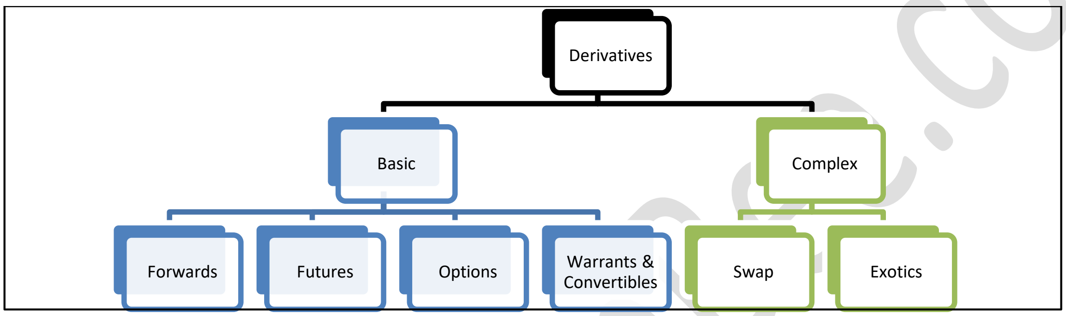
Feature	Description
Definition	Organization converts debtors into upfront cash by delegating collection to a third party or selling debtor invoice to a financier.
Types	With Recourse: Factor can collect unpaid invoice from client. Without Recourse: Risk of non-payment falls on factor.
Difference from Bills Discounting	Factoring involves financial institutions (factors) providing services beyond financing, such as credit investigation and debtor management.

Derivatives

Meaning of Financial Derivatives

- A derivative is a **financial instrument** whose value is **derived from** the value of another asset, which is known as the **underlying asset**.
- The underlying asset can be any asset like Stock or equity share, Bond or debenture etc.
- Financial derivatives may be traded **over-the-counter (OTC) or exchange-traded.**

Types of Financial Derivatives



Box 1: Understanding some terms

- **Spot price** = it is the current market price of an asset at which it can be bought/sold as of now in the spot market
- **Future price/expiration price** = the price of the asset at the specified date in future or on the date when the derivative contract will expire
- **Long position/going long** = it means to be in a buying position or agreeing to buy an asset
- **Short position/going short** = it means to be in a selling position or agreeing to sell an asset
- **Short sell** = it means to agree to sell an asset which one does not own as of now
- **Squaring off =** taking an opposite position to the original one so as to nullify the effect of the contract. For example, if a person has bought a derivative contract, he may sell another derivative contract with same terms and is said to have squared off his position

Forwards Contract

- A contract between **two parties**, where one party **agrees to buy** and other party **agrees to sell**, a **specified quantity** of an asset at the **specified price** at a **specific date** in future.

- **Example:** A sugar farmer, currently planting sugarcane at a market price of Rs. 110 per kg, anticipates a price drop in six months when his crop is ready for sale. To mitigate this risk, he enters into a forward contract with a sugar company, agreeing to sell 500 kg of sugarcane at a future price of Rs. 100 per kg, regardless of market fluctuations.
- The profit or loss position of the buyer and seller of the contract changes with the change in spot price of the underlying asset as follows:

Price relation	Buyer (Long position)	Seller (short position)
Spot price > forwards price	Profit	Loss
Spot price < forwards price	Loss	Profit

Futures Contract

- Futures contract is an exchange-traded contract to buy or sell an underlying asset at a future date at an agreed price.
- **Example:** Arvind, anticipating a price rise for ABC Ltd's shares in three months, purchases a futures contract on the National Stock Exchange. This contract represents 100 equity shares of ABC Ltd, allowing him to potentially profit from the expected increase in share price.
- In the futures market, both parties involved in a futures contract must open a margin account with a clearing house and deposit a specified amount of money, known as a margin. This requirement ensures that both parties have financial security and commitment to fulfill the contract.
  - **Initial Margin:** The amount required to open a futures position is called an initial margin.
  - **Maintenance Margin:** This is the minimum margin account balance required to retain the futures position.

Forwards and Futures Contract - comparison

Similarities

Parameter	Details
meaning	Contract to buy/sell an underlying asset in future at a pre-determined price
Underlying asset	Stock/equity, debenture/bond, commodity, currency, interest rates
Pay off	Unlimited profit or loss possible

Options

- An option is a *derivative* which gives the owner the **right but not the obligation** to sell/buy a specified quantity of the underlying asset at a particular price **on or before** a specified time.
- You pay a price to get this right to buy/sell which is called **option premium**.
- Options contract is also an **exchange-traded derivative**.
- There are two types of options:

- **Call Option** – Buyer has the right but not obligation to **BUY**.
- **Put Option** – Buyer has the right but not obligation to **SELL**.

- The two parties of option contract:
  1. Buyer of option (known as **option holder**)
  2. Seller of option (known as **option writer**)
- Depending on when an option can be exercised, an option can be:
  1. **American Style Option** –can be exercised at any time up to the expiration date. Options on individual securities available at NSE are American style options.
  2. **European Style Options** - can be exercised only on the expiration date. All index options traded at NSE are European Options.

- **Concept of ITM, OTM & ATM**

Money-ness	Call Option (i.e. option to Buy)	Put Option (i.e. option to Sell)
In the Money *(i.e. profit and hence option will be exercised)*	Spot Price > Strike Price	Spot Price < Strike Price
At the Money *(indifferent)*	Spot Price = Strike Price	
Out of the Money *(i.e. loss and hence option will not be exercised)*	Spot Price < Strike Price	Spot Price > Strike Price

- Animesh, who is optimistic about ITC Ltd's upcoming quarterly results, expects the company's profit to exceed market estimates. However, uncertain of the exact outcome, he opts to buy a Call Option rather than purchasing the company's shares directly. Animesh buys one Call Option contract, representing 100 shares of ITC, with a three-month expiration. He pays Rs. 20 per option with a strike price of Rs. 250, while the current share price of ITC is Rs. 225. This strategy allows him to potentially benefit from a price rise with a limited investment risk.

Swaps Contract

- An **agreement** between two parties to **exchange one set of cash flows for another.**
- It is like multiple forward contracts over a period of time.
- These are traded **over the counter** (OTC) and are customized contracts
- **Types of Swaps:**
  1. **Interest rate swaps:**
     - In Interest rate swaps, only the interest rate related cash-flows are swapped between two parties on a notional principal amount.
     - Plain Vanilla Interest Rate Swap: Exchanges fixed for floating rate payments on a notional amount for a set period.
     - Basis Swap: Exchanges payments between two different floating rates, each tied to separate benchmarks.

2. **Currency swaps:** Exchanges principal and interest payments between two currencies to hedge against exchange rate fluctuations.
3. **Credit Default Swap (CDS):** A financial derivative allowing investors to transfer credit risk associated with a fixed income product to another investor. It typically requires ongoing premium payments, similar to an insurance policy.

Difference between various types of derivatives

Particulars	Forwards	Futures	Options	Swaps
Trading	OTC	On exchange	On exchange	OTC
Type of contract	customized	standardized	standardized	customized
Underlying asset	Equity, bonds, commodity, currency, interest rates, etc.	Equity, bonds, commodity, currency, interest rates, etc.	Equity, bonds, commodity, currency, interest rates, etc.	Currency and interest rates
Execution of contract	Must be executed	Must be executed	Option holder - right but no obligation to execute	Must be executed
Premium payment	No extra premium payment	No extra premium payment	Option premium to be paid	No extra premium payment (except in case of CDS)
Pay off	Unlimited profit or loss	Unlimited profit or loss	Option holder – unlimited profited but limited loss	Unlimited profit or loss
Counter party risk	Yes	No	No	Yes
Credit Risk	Yes	No	No	Yes

Participants in Derivatives Market

Hedgers

- uses derivatives as a means to reduce risk
- provide economic balance to the market

Speculator

- takes risk to make profit in derivative market
- provide depth and liquidity to the market

Arbitrageur

- simultaneously transacts in 2 or more markets to take advantage of price discrepancies in these markets
- bring price uniformity and discovery

Bonds- Meaning

A bond is a debt instrument in which an investor loans money to an entity that borrows the funds for a defined period of time at a variable or fixed interest rate (known as a coupon).

Features of a Bond

- **Issue Price:** The initial price at which bonds are sold, often equal to the Face Value for coupon-bearing bonds, but issued at a discount for Zero Coupon Bonds.
- **Face Value (Principal/Par Value):** The amount on which interest payments are calculated and is paid to the investor at maturity, sometimes with an additional premium.
- **Coupon/Interest:** Regular cash flow from the bond, paid on specific dates.
- **Coupon Frequency:** Regularity of interest payments, which can be monthly, quarterly, semi-annually, or annually.
- **Maturity Date:** The future date when the principal is repaid and the bond expires.
- **Redemption Value:** The amount paid to the bondholder at maturity.
- **Duration:** Measures bond price sensitivity to interest rate changes, reflecting rate risk rather than time until maturity.
- **Sinking Fund Provision:** Requires the issuer to save funds periodically to repay the bond at maturity.
- **Credit Quality:** Indicates the risk of default; higher risk (junk bonds) results in higher interest, while lower risk (investment-grade bonds) yields lower interest.
- **Call Option:** Allows the issuer to repay the bond early, ceasing interest payments, which can disadvantage investors due to reinvestment risks.
- **Put Option:** Gives investors the right to demand early repayment.
- **Bond Indenture:** The contract detailing the above terms between issuer and bondholder.

Types of Bonds

On the basis of coupon-payments

- **Plain Vanilla Bonds:** Simple bonds with a fixed coupon rate, specific maturity, and typically redeemed at face value.
- **Zero Coupon Bonds:** Issued at a discount (e.g., Rs 960) and redeemed at face value (e.g., Rs 1000) at maturity, with the difference representing the investor's yield.
- **Deferred-Coupon Bonds:** Initially no coupon payments; higher coupons start after initial years to compensate. Common among businesses with delayed revenue generation.
- **Step-Up Bonds:** Coupons increase gradually over time, suited for entities expecting phased revenue growth.
- **Floating Rate Bonds:** Coupons adjust according to a benchmark rate (e.g., repo rate plus a spread).

- **Inverse Floaters:** Coupon rates move inversely to a benchmark interest rate, issued by corporates or governments.
- **Participatory Bonds:** Fixed rate with potential coupon increases if issuer's income exceeds a pre-set level, and reductions when income falls.
- **Income Bonds:** Similar to Participatory bonds but coupon rates do not decrease even if the issuer's income drops.
- **Payment-in-Kind Bonds:** Interest paid with additional bonds rather than cash, effectively increasing debt.
- **Inflation Indexed Bonds:** Principal and coupon payments are adjusted for inflation, preserving the bond's real value.

<u>On the basis of Redemption of the bond</u>:

- **Perpetual Bonds:** Pay fixed coupons indefinitely, with bonds over 100 years often classified as perpetual.
- **Callable Bonds:** Issuer can repay before maturity at a pre-agreed price; typically have higher coupon rates.
- **Puttable Bonds:** Holders can demand early repayment; generally offer lower coupon rates than straight bonds.
- **Serial Bonds:** Portions mature annually, reducing total debt over time; suited for projects with consistent revenue streams.

<u>On the basis of Convertibility</u>

- **Convertible Bonds:** Bonds that can be converted into equity shares at a pre-set time and price.
- **Foreign Currency Convertible Bonds (FCCBs):** Subscribed by non-residents, these bonds are denominated in foreign currency, carry fixed interest, and can convert into equity or GDRs, generally maturing in 5+ years.
- **Foreign Currency Exchangeable Bonds (FCEBs):** Issued by Indian companies in foreign currency, exchangeable into equity shares of another company within the same promoter group, subject to RBI approval and specific eligibility criteria.
- **Non-Convertible Debentures (NCDs):** Long-term debt instruments that cannot be converted into stock, offering regular interest payments and available as secured or unsecured, with safety and risk rated by credit agencies.

<u>Other Types of Bonds</u>:

- **Yankee Bonds:** Dollar-denominated bonds issued in the US by non-US entities.
- **Samurai Bonds:** Yen-denominated bonds issued in Japan by non-Japanese issuers.
- **Shogun Bonds:** Non-yen denominated bonds issued in Japan by foreign entities.
- **Eurobonds:** International bonds issued in a currency other than the issuer's domestic currency.
- **Masala Bonds:** Rupee-denominated bonds issued outside India by Indian entities to fund projects.

- **Green Bonds:** Bonds issued to finance environmentally beneficial projects, also known as climate bonds.

Difference between Bond and a Debenture

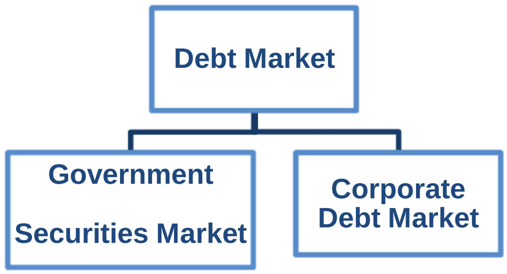
	Bonds	Debentures
Security	Bonds are almost certainly issued by entities backed by collateral	Debentures can be secured or unsecured debts, but normally are issued without collateral.
Issuers	Bond issuers are mostly Government and large financial institutes, which makes bonds the most secured debt instruments.	Debentures are mostly issued by private Companies.
Rate of Returns	As deemed more secured debt instruments, bonds offer lower returns than debentures	Companies often issue debentures to fund business expansion or a particular project making debts riskier. Therefore, the rates offered are higher.
Repayment Terms	As bond issuers are mostly Government-backed institutes, the repayment doesn't get affected by issuer performance.	Returns of debentures do depend on issuer performance. Particularly in the case of convertible debentures, project success plays a key role in repayment terms.
Liquidity	They are less liquid as vast secondary market is not available	Debentures offer higher liquidity in the secondary markets due to the convertibility feature.
Payment on liquidation of the firm/company	In the unlikely event of bankruptcy of these debt instruments, bondholders take priority before debenture holders	Debenture holders take priority before other shareholders

Corporate Debt Market

- A marketplace where fixed-income securities that offer fixed returns at regular intervals are traded.
- These securities provide consistent interest payments and principal repayment at maturity, akin to bank loans but with tradable instruments issued by borrowers, allowing investors to exit before maturity.

Types of Debt Market

A Debt Market can be broadly classified into:-



Government Securities Market:

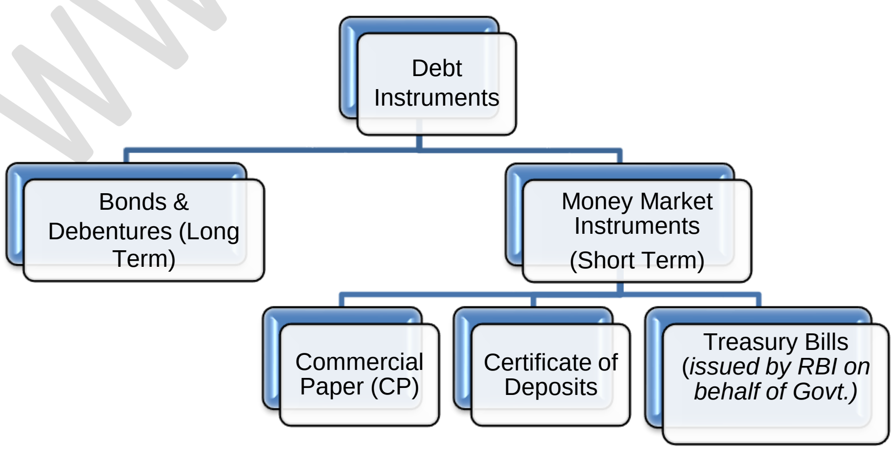
- A market for bonds and securities issued by various government bodies and agencies, **regulated by the RBI.**
- Major participants include commercial banks, primary dealers, institutional investors, and now retail investors through non-competitive bids.
- Foreign Portfolio Investors can participate within set limits.

Corporate Debt Market:

- A market for fixed-income securities issued by corporates, regulated by SEBI. Participants include corporates, institutional investors, FPIs, and retail investors.

Types of Debt instrument

Debt instruments issued are Bonds, Government Bonds, Debentures, Treasury Bills, Certificate of Deposits, Commercial Papers, etc.



A) Short Term Financing Instruments (Period upto 1 year)

- **Commercial Paper (CP):** Unsecured, short-term promissory note issued by corporates and financial institutions for 7 days to 1 year to meet immediate capital needs.
- **Certificate of Deposit (CD):** Negotiable, dematerialized financial instrument issued by banks and select financial institutions for fixed short-term durations.

B) Long Term Financing Instruments (Period more than 1 year):

- **Fixed Rate Bonds:** Popular, offer fixed semiannual coupons and principal repayment at maturity.
- **Floating Rate Bonds:** Coupons adjust based on benchmarks like MIBOR, with caps and floors to limit rate changes.
- **Debentures:** Secured, fixed-interest debt instruments, available in convertible and non-convertible forms.
- **Zero Coupon Bonds:** Issued at a discount, pay full face value at maturity without periodic coupons.
- **Cash Management Bills (CMBs):** Short-term, issued by the government through RBI to manage temporary cash needs.
- **Market Linked Debentures (MLD):** Returns tied to market indices, available in principal-protected and non-protected forms.

Structure of Corporate Debt Market

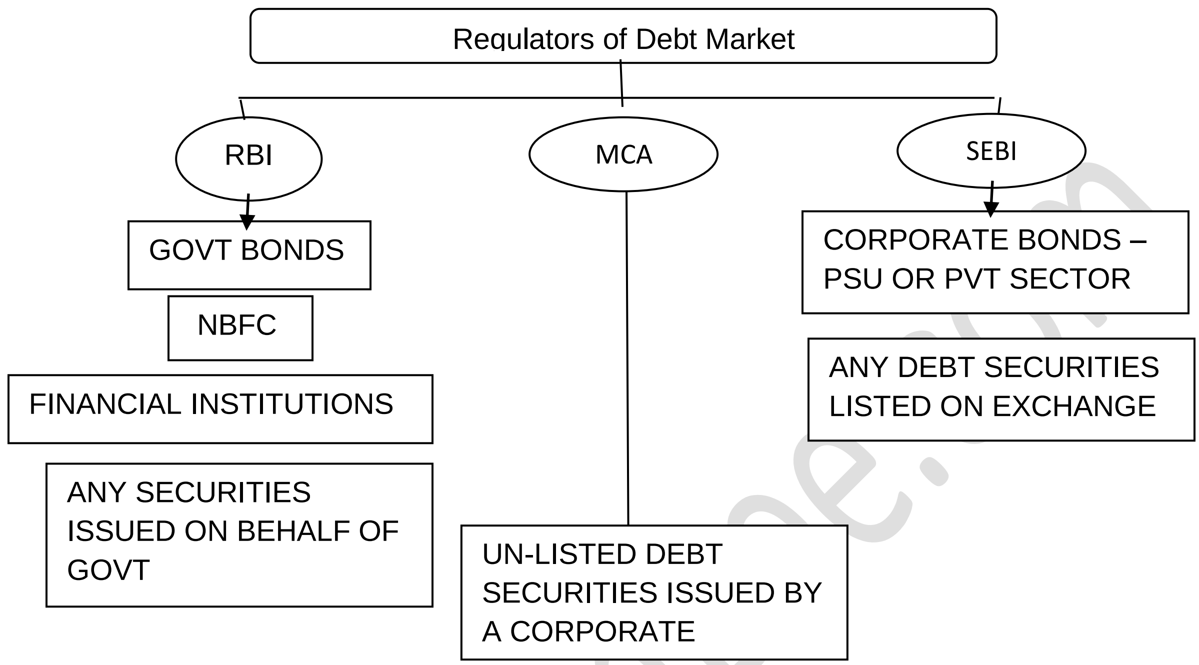
This Debt Market can be classified into

- **Primary Market:** Corporates issue debt securities like bonds and commercial papers for the first time, either through public prospectus or private placement.
- **Secondary Market:** Existing corporate debt securities are traded on platforms like NSE & BSE's Wholesale Debt Segment.

Securities can be issued in two ways:

- **Public Issue:** Bonds are offered to the general market, involving higher borrowing costs.
- **Private Placement:** Bonds are issued directly to a few institutions, a prevalent and cost-effective method in India.

Regulatory Framework:



Basics of Bond Valuation

- The various long-term sources of finance include:

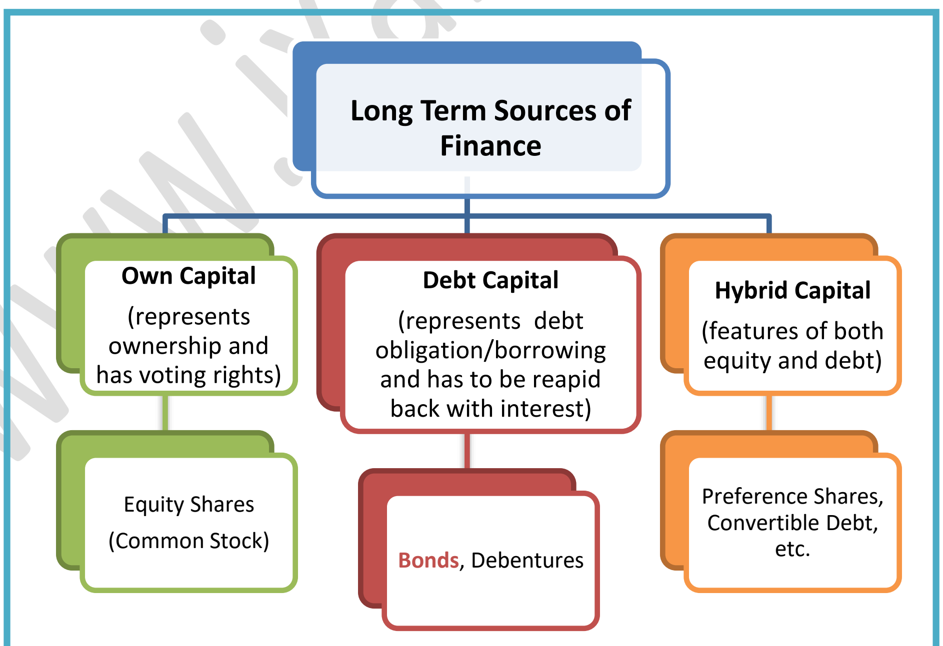
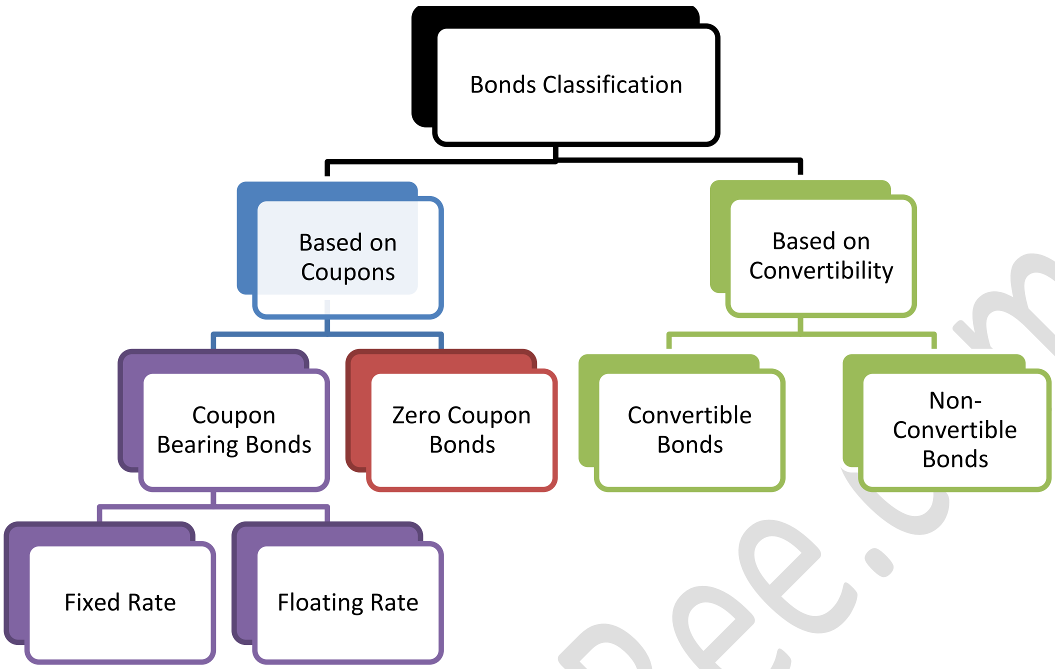


Figure 1: Long term sources of Finance



Concept of Yield

Yield is the return generated from a bond investment, crucial for assessing returns and comparing investment opportunities; it varies based on the investment and exit timings.

Yield-to-Maturity (YTM)	Current Yield
• return earned by holding the bond till its maturity. • the earnings would include all remaining coupon payments and the redemption value • YTM is also used as a proxy for discount rate/opportunity rate/opportunity cost of capital/required rate of return.	• return earned by purchaing the bond in market and selling it before maturity. • the earnings for bondholder includes the coupon only.

1. **Approximate YTM** = $\dfrac{Coupon\ amount\ periodically + \dfrac{(Redemption\ Price - Market\ Price)}{No\ of\ years\ left\ to\ maturity}}{\dfrac{(Redemption\ Price + Market\ Price)}{2}}$

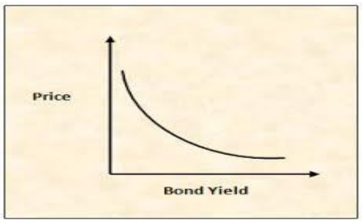
2. Current Yield = $\frac{Absolute\ Coupon\ Amount\ \ periodically}{Market\ Price}$

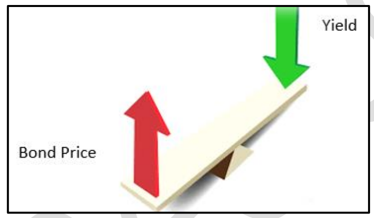
Value of the Bond

- The intrinsic worth of a bond to investors, equal to the present value of future cash flows, including periodic coupons and the redemption value at maturity.

Relationships between Interest rate and Bond price

- **Bond Prices and Interest Rate** (or also referred to as the YTM/Required rate of return/market rate) **are inversely related.**





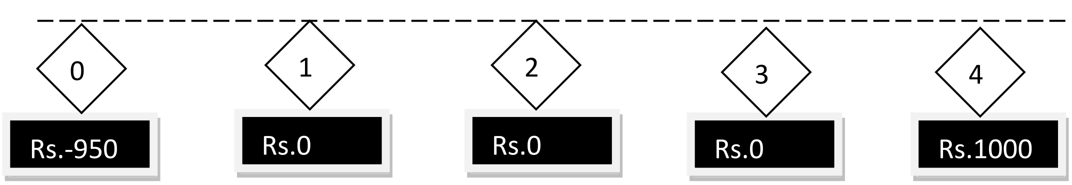
- When bond prices increase, returns decrease due to smaller capital gains. Conversely, if market yields rise, existing bonds lose appeal, increasing supply and reducing prices, as higher yields decrease bond values when discounting future cash flows.

Relationship between YTM, Coupon and Bond Prices

YTM vs. Coupon	Bond	Bond Price vs. FV of Bond	Price movement till maturity
YTM > Coupon Rate	Discount Bond	Price < FV	Bond Price will increase to par over its maturity
YTM < Coupon Rate	Premium Bond	Price > FV	Bond Price will decrease to par over its maturity
YTM = Coupon Rate	Bond at Par	Price = FV	Bond Price is constant over the life of the bond

Zero Coupon Bonds

A bond that pays no periodic interest, sold at a discount to face value, and known as Deep Discount Bonds. The implicit interest is the difference between the issue price and the redemption value. For example, a bond with a 4-year maturity and a face value of Rs.1000 issued at a Rs.50 discount results in a redemption cash flow of Rs.1000.



Types of Zero-Coupon Bonds in India

- **Treasury Bills (T-Bills):** Short-term government-issued money market instruments, available in 91, 182, and 364-day tenors, issued at a discount and redeemed at face value, e.g., a 91-day T-Bill with a face value of Rs.100 may be issued at Rs.98.20.
- **Cash Management Bills (CMBs):** Similar to T-Bills, these are short-term instruments issued by the central government to manage temporary cash needs, issued for maturities under 91 days.
- **STRIPS:** Securities derived from G-Secs, separating each coupon payment and the principal into individual securities, traded separately from the original bond and not issued via auctions.

Difference– Zero Coupon Bonds and Coupon Bonds

Factor	Zero Coupon Bonds	Coupon Bonds
Meaning	Bond that pays no periodic interest and are sold at a discount to par value.	Bond that pays periodic interest in the form of coupons
No. of Cash Inflows	1	>=1
Reinvestment and Price Risk	• No intermediate Cash Inflows, therefore No-Reinvestment Risk • Price can change as per the demand and supply, therefore Price Risk is present	• Intermediate Cash Inflows, therefore Reinvestment Risk is present • Price can change as per the demand and supply, therefore Price Risk is present
Yield to Maturity	$\sqrt[n]{\frac{Redemption\ Value}{Issue\ Price}} - 1$	Actual YTM calculated through trial and error method which equates the PV of future inflows) with the issue price
Duration	Same as maturity of the ZCB	Less than maturity of the bond

Various terms used for Interest rates:

- **Treasury Rates:** Risk-free rates for government debt instruments like Treasury bills and bonds, assumed secure as they are in the government's currency.
- **Repo and Reverse Repo Rates:** Interest rates from repurchase agreements, where securities are sold and later repurchased, used for short-term funding with minimal credit risk.
- **Call Rates:** Rates in the unsecured short-term loan market, similar to overnight rates in mature markets.
- **LIBOR (London Interbank Offered Rate):** Benchmark rate reflecting inter-bank deposit rates, quoted daily in major currencies.

- **MIBOR (Mumbai Interbank Offered Rate):** LIBOR equivalent for the Mumbai inter-bank market, influencing floating rate instruments.
- **MIFOR (Mumbai Inter Bank Forward Offer Rate):** A benchmark combining LIBOR and forward premiums, adjusting post-LIBOR era for derivative pricing.
- **Zero Rates:** Interest for n-year investments paid at maturity, known as spot or zero coupon rates, reflecting pure interest earnings without interim payments.
- **Forward Rates:** Projected future earnings from bonds, calculated using spot rates or yield curves, used to hedge against interest rate fluctuations.
- **Risk-Free Rate:** Used in derivatives evaluation, generally derived from treasury securities, deemed low-risk due to high demand and regulatory treatments.

Risks associated with a Bond

Macaulay's Duration – The Conceptual Understanding

- Represents the weighted average time it takes to recover the initial bond investment through its cash flows, considering both coupon payments and principal repayment.
- Duration, different from maturity, is typically shorter and influenced by coupon rates—higher rates lead to shorter durations.

Relationship between Duration (D), Maturity, Coupon and Yield

Direct relation between Duration and Maturity
Duration increases with the maturity of the bond. For example, if the 2 bonds A and B, offering 8% coupon having maturity of 2 and 5 years respectively, then the duration of bond B we will be more than that of bond A.

Inverse relation between Duration and market Yield
Duration decreases as yield increases. For example if the yield increases from 8% to 9%, the duration of the bond will decrease.

Inverse relation between Duration and Coupons
Duration decreases as the coupon rate increases. For example, if the 2 bonds A and B having same maturity, offering 8% and 6% coupon respectively, then the duration of bond A we will be less than that of bond A.

Duration with no coupons
In case of **zero coupon bonds**, **duration** of the bond will be **equal to the bond maturity**. For example, duration of a bond having a face value of Rs. 1000, issued at 970 and having a maturity of 5 years will be equal to 5 years.

Modified Duration (MD)

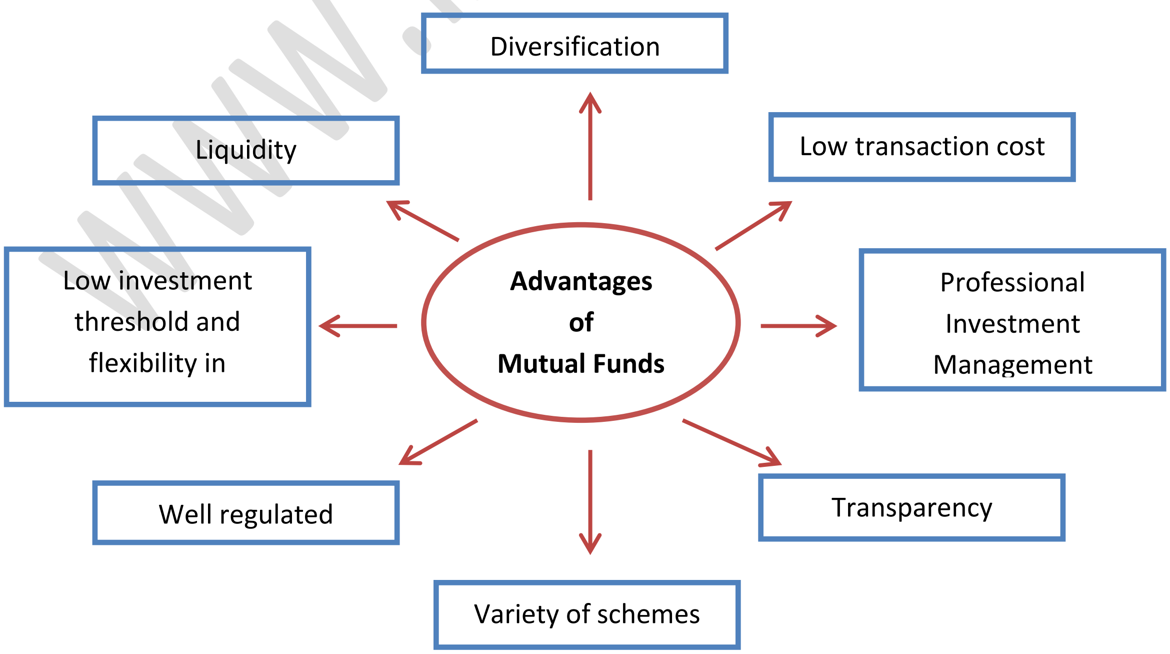
Measures a bond's price sensitivity to interest rate changes, indicating the percentage change in bond value per 1% change in interest rates. It's calculated using Macaulay Duration and adjusts for the market interest rate, making it crucial for assessing interest rate risk.

***Note: Here topics are covered in a summarized manner for revision purpose, for comprehensive understanding please refer to detailed notes.**

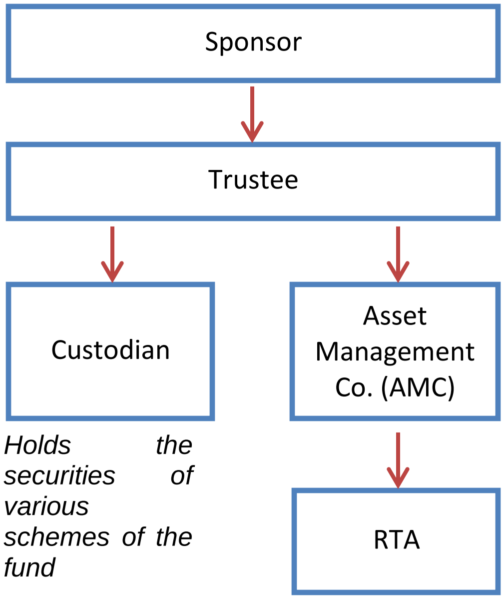
Mutual Funds - Meaning

- A mutual fund pools money from investors to purchase securities, diversifying risk across various industries and sectors.
- Investors receive units proportional to their investment and share profits or losses accordingly. Mutual funds offer various schemes with distinct objectives and must register with SEBI, which regulates them under the SEBI (Mutual Funds) Regulation, 1996.

Advantages of Mutual Funds



Structure of Mutual Funds



Floats the mutual fund like a promoter of a company; contributes ***at least 40% of the net-worth*** *to set up the mutual fund*

Holds the assets and property of the fund on behalf of the unit-holders; vested with the general power of superintendence and direction over AMC; makes sure that the fund runs as per the SEBI guidelines

AMC approved by SEBI, manages the funds by making investments in various types of securities by appointing a Fund Manager

Holds the securities of various schemes of the fund

- **Board of Trustees:** Manages the mutual fund's assets on behalf of unit-holders with a mandate that two-thirds must be independent of the sponsoring company, ensuring the protection of investor interests.
- **Asset Management Company (AMC):** Fund managers responsible for investing and managing the pooled funds in various securities based on market research, while handling administrative tasks and charging a management fee. At least 50% of AMC directors must be independent.
- **Custodian:** A bank approved and registered under SEBI regulations that safeguards the mutual fund's portfolio securities, ensuring legal compliance and security of investments.

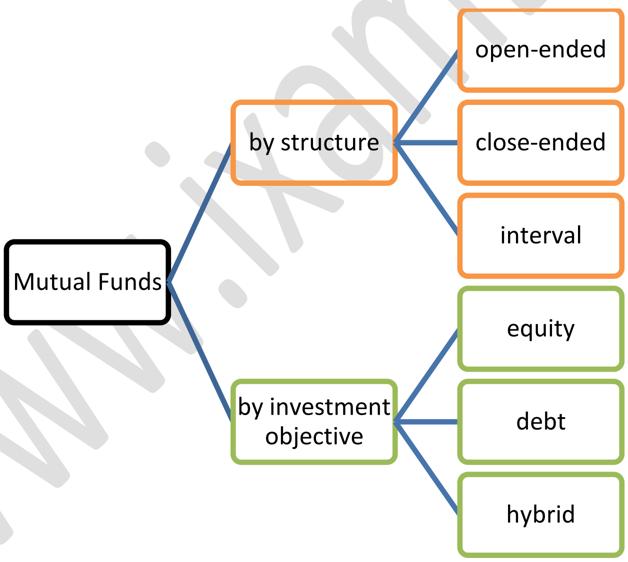
Mutual Fund Basic Terms & Concepts

- **Net Asset Value (NAV):**
    - Reflects the market value of a mutual fund's securities minus liabilities, divided by the number of units.
    - NAV fluctuates daily with market conditions and must be reported daily on the AMFI and fund's websites by 9 p.m.
- **Assets Under Management (AUM):**
    - The total market value of the securities a mutual fund holds, representing the scale of the fund's operations.
- **Load:**

- Fees charged to mutual fund investors, including management fees for portfolio management, entry loads (abolished in India since August 2009), and exit loads for unit redemption.
- A no-load fund charges neither entry nor exit fees.

- **Total Expense Ratio (TER):**
  - Measures the total costs of managing a mutual fund, including management fees, operational costs, and distributor commissions, expressed as a percentage of total assets.
  - SEBI regulates the structure of TER, including base TER rates and potential additional charges based on specific conditions like inflows from certain cities.
  - Changes in TER must be disclosed to investors in advance and reported daily on AMFI and AMC websites.
- **Systematic Investment Plan (SIP):**
  - Allows regular, periodic investments into a mutual fund, leveraging the benefits of dollar-cost averaging and disciplined investing.
  - It reduces the impact of market volatility by spreading purchases over time, as illustrated by an example where an investor accumulates units at varying NAVs, resulting in an average cost per unit.

Types of Mutual Funds

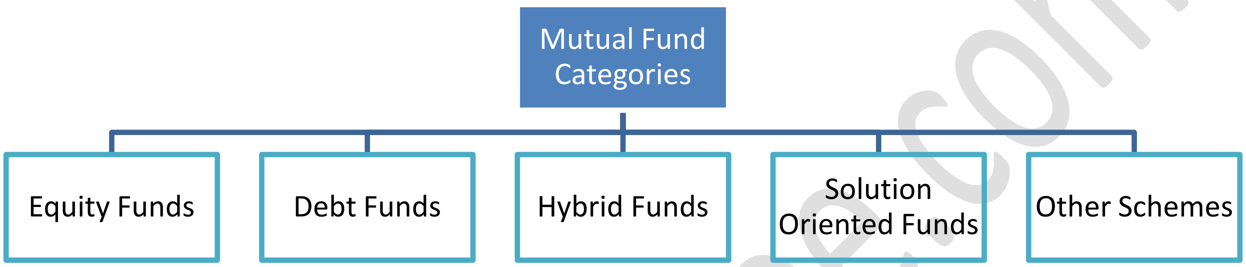


Types of Mutual Funds – based on Structure:

- **Open-ended Mutual Funds:** Continuously open for subscription and redemption at daily NAV, providing high liquidity. Investors can exit the fund at any time as the AMC is required to buy back the units at the daily declared NAV.

- **Closed-ended Mutual Funds:** Available only during the New Fund Offer (NFO) period with a fixed maturity (e.g., 3-5 years); units can't be bought post-NFO and are listed on exchanges for trading to offer exit options before maturity.
- **Interval Funds:** Allow transactions only during specific intervals at prevailing NAV, often listed on exchanges, and typically invest in debt, resembling Fixed Maturity Plans with investments locked until maturity for better fund management.

Types of Mutual Funds – based on investment objective



Equity Funds

- These funds primarily invest in equities, carrying higher risks with the aim of capital appreciation over the medium to long term.
- They offer various options like dividend payout or growth, and allow investors to switch options later as per their preferences.

SEBI has set a clear classification as to what is a large cap, mid cap and small cap:

Market Cap	Meaning
Large Cap	1st to 100th company in terms of full market capitalization
Mid-cap	101st to 250th company in terms of full market capitalization
Small cap	251st company onwards in terms of full market capitalization

Debt Funds

- Aimed at providing regular, steady income and capital preservation, these funds invest in fixed income securities like bonds, debentures, government securities, and money market instruments.
- They are less risky than equity funds but offer limited capital appreciation. NAVs of debt funds fluctuate with interest rate changes, increasing when rates fall and decreasing when they rise, though long-term investors may not be significantly affected by these fluctuations.

Hybrid Funds

- Designed to offer both growth and regular income, these funds invest in a mix of equities and fixed income securities as specified in their offer documents.
- Suitable for investors seeking moderate growth, hybrid funds typically exhibit less NAV volatility compared to pure equity funds.

***Note: For the subclassification of all please refer to the detailed notes.**

Solution Oriented Funds

S No	Type of Scheme	Meaning
1	Retirement Fund	This is a retirement solution oriented scheme that will have a lock-in of five years or till the age of retirement.
2	Children's Funds	This is children oriented scheme having a lock-on for five years or until the child attains the age of majority, whichever is earlier

Other Funds

Index Fund:

- Mirrors a market index (like Nifty or Sensex) by holding shares in the same proportion as the index.
- Passively managed, typically invests at least 95% of its assets in the securities of the specified index.

Fund of Funds (FoF):

- A mutual fund that invests in other mutual funds, suitable for investors who prefer managing one fund instead of multiple.
- Can diversify across various mutual fund schemes at a lower investment size, investing a minimum of 95% of its assets in underlying funds.

International Funds:

- Invests in securities outside India or into master funds domiciled abroad, covering various regions like emerging or developed markets.
- Examples include funds focused on specific sectors like gold or commodities.

Gold Funds:

- Invests in gold ETFs, providing an alternative to direct stock exchange purchases without requiring a broking account.

Exchange Traded Fund (ETF):

- Trades on stock exchanges like stocks, offering liquidity and price changes throughout the trading day.

- Requires investors to have demat and trading accounts as ETFs are bought and sold through the stock exchange.

Capital Protection-Oriented Scheme:

- A hybrid fund mainly invests in fixed-income securities to preserve capital, with a smaller portion in equities for growth.
- Close-ended with a fixed maturity, structured to ensure the principal amount is preserved until maturity without any guarantee of returns.

Capital Asset Pricing Model (CAPM)

Equity Expectations for Return:

- Risk-free assets like government securities (G-secs) provide a baseline return with no credit risk.
- For example, the return on a 10-year G-sec, considered the risk-free rate, is assumed to be 8% per annum.
- Equity investors, facing higher risks, expect a "Risk Premium" as compensation for additional organizational risks.

Systematic and Unsystematic Risks

SYSTEMATIC RISK

- The risks that are common to the entire system, such as changes in Inflation, Repo Rate, Market sentiments
- The entire system gets effected (Macro nature)
- Every firm is exposed to these risks being a part of the entire economic and market system, as such this risk is **un-diversifiable**

UNSYSTEMATIC RISKS

- These risks arise because of micro factors of an organisation such as operating risks, or the high financial burden
- For Example, if a regulation has come out by the government for sugar sector, it will have an effect to the companies operating in the sugar industry
- This risk is **diversifiable,** as such

CAPM (Capital Asset Pricing Model) - Components

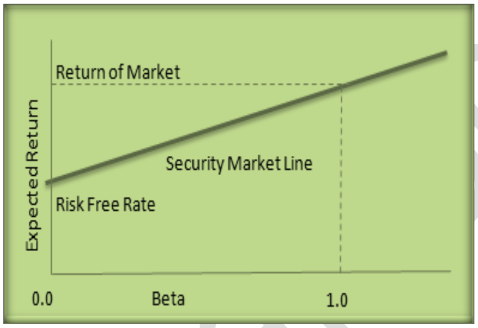
- Risk free rate of return
- Risk Premium

- For undertaking risks in the stock, they want premium so as to get a compensation for additional amount of risks they are undertaking.

- Beta – Market Risk
  - CAPM gives compensation for undertaking only the market risks , thus, Risks Premium = $\beta*(R_m - R_f)$, where,
    R_f – Risks free rate of return
    R_m – Return of the market

CAPM – Main Equation and Calculation of Cost of Equity

- Expected return on a stock/ Cost of Equity is: $E(R) = R_f + \beta*(R_m - R_f)$



Stock Beta	Nature of Stock	Meaning	Return with respect to Market
Beta (β) < 1	Defensive	Asset is less volatile i.e. lesser risk (relative to the market)	Stock return will be less than market return
Beta (β) = 1	Neutral	Assets volatility is the same as the market	Stock return will be same as market return
Beta (β) > 1	Aggressive	Asset is more volatile i.e. higher risk (relative to the market)	Stock return will be more than market return

Concept of Alpha:

- Alpha represents the excess return of an investment relative to a benchmark index.
- According to the CAPM model, alpha is the difference between the actual return and the required rate of return.
- Alpha can be positive (indicating outperformance) or negative (indicating underperformance compared to the market portfolio).